

FACTSHEET FS21 · Machine-learning portfolios: hierarchical risk parity vs equal weights

What changed on 2 October 2026. Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS_RERUN_2026-10-02.md.

Does clustering stocks with a machine-learning tree give a better portfolio than splitting the money equally?

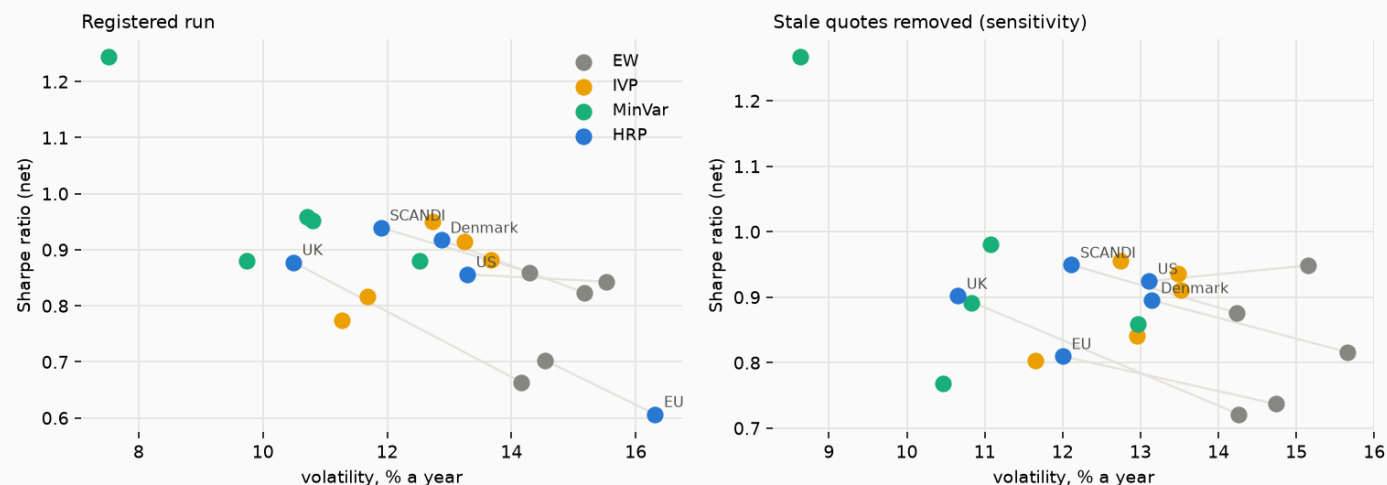
Henrik Gjerning · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · January 2013 – September 2026

Key facts

Pre-registration	planning/PREREG_FS21.md, written 30 Sep 2026 before any FS21 number; one deviation (a stale-quote filter), reported as a sensitivity
Universes	US (Sharadar, S&P 500 members), EU, UK, Denmark, SCANDI point-in-time universes; World left out (its USD conversion files are not in the folder)
Portfolios	Equal weight (EW); inverse variance (IVP); long-only minimum variance with Ledoit–Wolf shrinkage (MinVar, cap 5–10%); hierarchical risk parity (HRP, López de Prado 2016)
Rebalancing	monthly, 252 trading days of history, long only, fully invested; costs 10 bp per unit of one-way turnover (25 bp sensitivity)
Tests	T1 HRP beats EW, T2 HRP beats IVP: pooled Sharpe difference over five markets, joint block bootstrap, one-sided $p < 0.025$

In one paragraph. Neither test passes. Pooled over five markets HRP's Sharpe ratio is +0.06 above equal weight (p 0.18) and -0.03 against simple inverse-variance weights (p 0.82). HRP does what it is built to do, cutting volatility from about 15% to 13% a year, but most of that comes from tilting towards quiet stocks, which inverse-variance weights do without any tree. In the EU the registered run exposed a data problem: a few **stale Yahoo quotes** (prices that never move) look riskless, so HRP put on average 42% of the EU book into a single stock (IVP 29%). With stale series removed (a deviation, reported as a sensitivity) HRP beats EW by +0.08 Sharpe (p 0.07) and IVP by +0.01 (p 0.43): still not significant. The plain long-only minimum-variance portfolio had the best average Sharpe (0.98 vs 0.78 for EW). The lesson matches FS14 and DeMiguel, Garlappi & Uppal (2009): the machine-learning tree is elegant, but on real stocks the gain over 1/N comes from owning less risky stocks, not from the clustering.

Four ways to weight the same stocks, 2013–2026: grey lines join equal weight (EW) to HRP in each market



1. Market by market

Market	Method	Sharpe (registered)	Sharpe (stale removed)	Volatility	Max drawdown	Effective no. of stocks	Largest weight (avg)	Turnover / month	Sharpe at 25 bp
US	EW	0.84	0.95	15.5%	-28%	501	0.2%	3%	0.84
	IVP	0.88	0.94	13.7%	-24%	377	0.9%	4%	0.87
	MinVar	0.96	0.98	10.7%	-15%	31	5.0%	22%	0.89
	HRP	0.86	0.92	13.3%	-22%	322	2.0%	12%	0.82
EU	EW	0.70	0.74	14.5%	-28%	256	0.4%	4%	0.69
	IVP	0.82	0.84	11.7%	-25%	86	27.8%	8%	0.79
	MinVar	0.88	0.77	9.7%	-19%	28	5.0%	16%	0.82
	HRP	0.61	0.81	16.3%	-23%	38	42.5%	13%	0.58
UK	EW	0.66	0.72	14.2%	-30%	310	0.3%	4%	0.65
	IVP	0.77	0.80	11.3%	-24%	130	4.6%	5%	0.76
	MinVar	1.24	1.27	7.5%	-16%	29	5.0%	13%	1.18
	HRP	0.88	0.90	10.5%	-23%	65	9.6%	13%	0.83
Denmark	EW	0.82	0.82	15.2%	-33%	22	4.6%	4%	0.81
	IVP	0.91	0.91	13.2%	-27%	17	10.8%	5%	0.90
	MinVar	0.88	0.86	12.5%	-18%	12	10.0%	8%	0.86
	HRP	0.92	0.90	12.9%	-24%	16	12.7%	11%	0.89
SCANDI	EW	0.86	0.88	14.3%	-29%	71	1.4%	5%	0.85
	IVP	0.95	0.96	12.7%	-23%	57	3.8%	5%	0.94
	MinVar	0.95	0.89	10.8%	-14%	16	9.9%	14%	0.91
	HRP	0.94	0.95	11.9%	-22%	48	5.4%	12%	0.90

Net of 10 bp costs; Sharpe on returns without subtracting cash (local currency, as in the series' other long-only books). Effective number of stocks = $1/\Sigma w^2$, averaged over months.

2. Tests

Test	Registered	Stale quotes removed
T1 HRP – EW, pooled Sharpe	+0.061, p 0.181, 95% CI [-0.05, +0.19] → fail	+0.076, p 0.070 → fail
T2 HRP – IVP, pooled Sharpe	-0.028, p 0.823, 95% CI [-0.11, +0.04] → fail	+0.007, p 0.431 → fail

Per market (Ledoit–Wolf 2008 test of the Sharpe difference, HAC, reported only):

Market	HRP – EW (registered)	HRP – IVP (registered)	HRP – EW (stale removed)
US	+0.01 (t +0.24)	-0.03 (t -1.01)	-0.02 (t -0.42)
EU	-0.10 (t -0.55)	-0.21 (t -1.59)	+0.07 (t +1.24)
UK	+0.21 (t +2.63)	+0.10 (t +1.92)	+0.18 (t +2.20)
Denmark	+0.09 (t +1.19)	+0.00 (t +0.06)	+0.08 (t +1.04)
SCANDI	+0.08 (t +1.26)	-0.01 (t -0.23)	+0.07 (t +1.33)

3. What the tree does and does not add

- **Volatility:** HRP is 1.8% a year less volatile than EW on average, and IVP 2.2%. Most of the risk reduction is the inverse-variance tilt; the clustering adds little.
- **Concentration:** HRP holds the equivalent of far fewer stocks than EW, and in thin or dirty data it can concentrate heavily (EU). That fragility is a known criticism of risk-based weights: an estimation error in one variance becomes a big position.
- **Turnover:** 10–14% of the book a month for HRP against 3–5% for EW; costs matter at 25 bp but do not change the ranking.
- **Minimum variance:** the unglamorous optimiser with shrinkage and a weight cap did best in the UK and on average, consistent with the low-risk anomaly (FS04, FS10).

4. Caveats

- Local-currency, long-only, large and mid caps; 2013–2026 only.
- The stale-quote filter (more than 20% zero daily returns in the 252-day window) was added after seeing the EU result and is reported only as a sensitivity; the registered test is the verdict.
- Stocks without a return in a month contribute 0 (delisting returns may be missing), identical for all four methods.
- FS21 adds two trials to the ledger.

5. References

- DeMiguel, V., Garlappi, L. & Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the 1/N portfolio strategy? *Review of Financial Studies* 22(5), 1915–1953.
- Ledoit, O. & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis* 88(2), 365–411.
- Ledoit, O. & Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance* 15(5), 850–859.
- López de Prado, M. (2016). Building diversified portfolios that outperform out of sample. *Journal of Portfolio Management* 42(4), 59–69.
- Raffinot, T. (2017). Hierarchical clustering-based asset allocation. *Journal of Portfolio Management* 44(2), 89–99.

6. Reproduce

planning/PREREG_FS21.md → code/fs21.py (registered) and code/fs21.py clean (sensitivity) → code/build_fs21.py. Uses code/data.py (point-in-time universes) and code/perf.py.